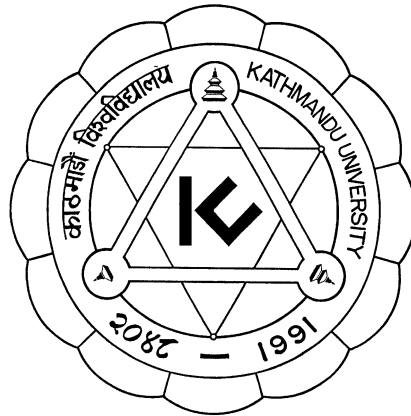


KATHMANDU UNIVERSITY

SCHOOL OF ENGINEERING

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

PROJECT REPORT



**CLOSED-LOOP OPERATION OF A BUCK CONVERTER
UNDER FLUCTUATING LOAD CONDITIONS**

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ABSTRACT

Efficient DC-DC power conversion is critical for modern electronic systems requiring stable voltage supplies. This project presents the design and simulation of a Buck Converter capable of regulating a 12V input to a stable 5V output. The system operates at a switching frequency of 5 kHz and utilizes a Pulse Width Modulation (PWM) switching approach. To mitigate error and improve response under varying load conditions, an analog Proportional-Integral (PI) controller is implemented using operational amplifiers. The compensator design is tuned to match the poles and zeros of the LC filter, ensuring system stability. The control signal is compared against a generated sawtooth waveform to dynamically adjust the duty cycle. This study validates the effectiveness of analog feedback loops in maintaining output voltage regulation despite load disturbances.

Keywords: PWM, Load Regulation, Analog Control, DC-DC Conversion.

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CHAPTER I: INTRODUCTION

The demand for efficient and stable DC-DC power conversion has become paramount. As electronic devices (portable smartphones to high-performance computing servers) require specific and tightly regulated voltage levels, the buck converter has emerged as a fundamental component. While a basic buck converter can step down voltage efficiently, its performance is often hindered by fluctuations in input voltage and varying load conditions. To address these instabilities, a control mechanism is required. By utilizing operational amplifiers to form a PI control loop, we can ensure that the output voltage remains constant despite external disturbances.

The Buck Converter

The Buck converter is a switch-mode DC-DC converter that steps down a high input voltage to a lower output voltage. It consists of four primary elements: Power MOSFET, Inductor, Capacitor, Freewheeling Diode.

Pulse Width Modulation (PWM)

PWM is the method used to control the power delivered to the load. By varying the duty cycle, the output voltage can be precisely adjusted. Here, it is done by comparing a control voltage with a high-frequency sawtooth or triangular wave.

Closed-Loop Control

A closed-loop system, however, uses a feedback loop to monitor the output voltage, compare it to a desired reference and generate an error signal. This signal is then processed by the controller to adjust the PWM duty cycle.

The PI Controller

The Proportional-Integral (PI) controller is a feedback mechanism that calculates the correction required based on the error signal:

Proportional Term (P): Provides an immediate response to the current error. A higher gain increases the speed of the system but can lead to instability.

Integral Term (I): Accumulates the error over time. Its primary function is to eliminate the steady-state error, ensuring the output voltage exactly matches the reference voltage over the long term.

Error Calculator

The error amplifier is an op-amp differentiation. It subtracts the feedback voltage from the reference voltage and creates an appropriate signal for the controller to process.

1.1 Objectives

- To design the power stage for a specific rating.
- To develop an analog PI control circuit using Op-Amps.
- To simulate the circuit in simulation software.

1.2 Problem Description

In an unregulated buck converter, any drop in the source voltage or an increase in the load current will cause the output voltage to drop. For sensitive electronic components, even a slight deviation can cause malfunctions. The challenge is to design an analog circuit that can sense these deviations and correct the duty cycle.

1.3 Limitations

The project has the following limitations:

- The desired output voltage for the converter is fixed and cannot be varied as per required.
- If the load voltage is above 5V, the controller breaks. This occurs due to the unavailability of a modulus op-amp circuit. The load voltage being above 5V sets the error as negative and thus the correction signal also goes to be negative.

CHAPTER II: TECHNOLOGY SURVEY

2.1 Components Required

Table 1: Components Required

S.N.	Name
1	LM 741
2	Inductor (1mH)
3	Capacitor (470uF)
4	IRFZ44N
5	LM 311
6	Resistors
7	Relay
8	Potentiometer

2.2 Formula used

- Output voltage:

$$V_o = D V_{in} \quad (1)$$

- Gain of inverting amplifier:

$$A_v = \frac{V_o}{V_{in}} = -\frac{R_f}{R_{in}} \quad (2)$$

- Output voltage of op-amp summer:

$$V_o = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \dots \right) \quad (3)$$

- Output voltage of op-amp integrator:

$$V_o(t) = -\frac{1}{RC} \int V_{in}(t) dt \quad (4)$$

Using the above formulas, we can obtain the relevant parameters within the circuit.

2.3 Calculations

Frequency: 5 Hz

Output Voltage: 5 V

Output Current: 2 A

Inductor Current Ripple: 30% (0.6 A).

Output Voltage Ripple: 50 mV (1% of output).

- Duty Cycle:

$$D = \frac{V_{out}}{V_{in}} = \frac{5V}{12V}$$
$$D \approx 0.417 \text{ (or 41.7\%)}$$

- Inductance (L):

We need to limit the current ripple to 0.6 A.

$$L = \frac{(V_{in} - V_{out}) \cdot D}{\Delta I_L \cdot f_{sw}}$$
$$L = \frac{(12V - 5V) \cdot 0.417}{0.6A \cdot 5000Hz}$$
$$L = \frac{7 \cdot 0.417}{3000}$$
$$L = \frac{2.919}{3000} \approx 0.000973 \text{ Henrys}$$

0.000973 H is rounded to a standard inductance of 1 mH.

- Capacitance (C):

We need to keep the voltage ripple under 50mV.

$$C_{out} = \frac{\Delta I_L}{8 \cdot f_{sw} \cdot \Delta V_{out}}$$
$$C_{out} = \frac{0.6A}{8 \cdot 5000Hz \cdot 0.05V}$$
$$C_{out} = \frac{0.6}{2000}$$
$$C_{out} = 0.0003 \text{ Farads}$$

Again, a standard capacitor with capacitance 470 uF is chosen.

- Differential Amplifier:

$$V_{out} = \frac{R2}{R5} \times (V_{non-inv} - V_{inv})$$

When $R2 = 1k\Omega$, $R5 = 1k\Omega$, $R3 = 1k\Omega$, $R4 = 1k\Omega$

$$\text{Gain} = \frac{1k\Omega}{1k\Omega} = 1$$

- Integration Unit :

The integration gain is given by

$$G = \frac{1}{R \times C}$$

$R9 = 10k\Omega$, $C = 100nF = 100 \times 10^{-9}F$

$$\text{Gain} = \frac{1}{10,000 \times 100 \times 10^{-9}} = \frac{1}{0.001} = 1000$$

- Proportional Unit:

$R6 = 4.7k\Omega$, $R7 = 1k\Omega$

$$A_v = -\frac{4.7k\Omega}{1k\Omega} = -4.7$$

- Summer :

$R10 = 1.1k\Omega$, $R11 = 1k\Omega$, $R12 = 1k\Omega$

$$\text{Gain} = -\frac{1k\Omega}{1.1k\Omega} = -0.9$$

CHAPTER III: METHODOLOGY

3.1 Block Diagram

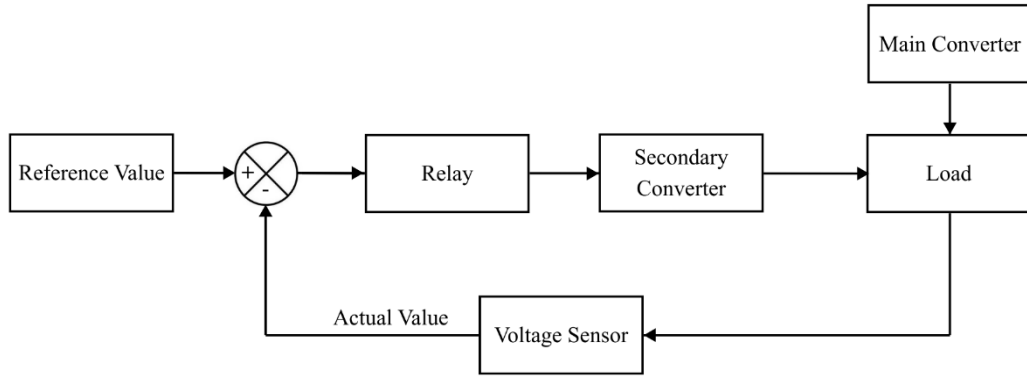


Figure 3.2: Block Diagram of the overall circuit

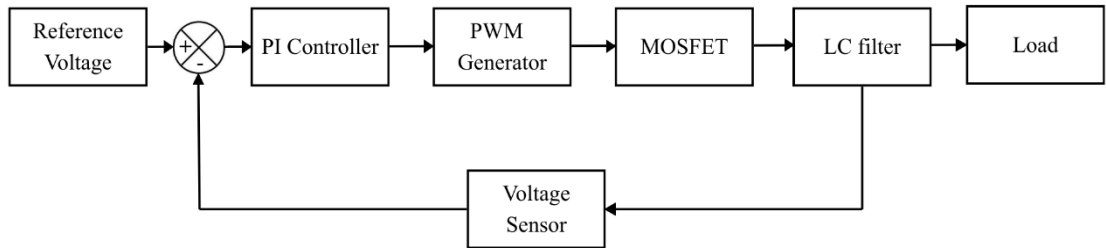


Figure 3.1: Block Diagram of the buck converter

Figure 3.1 is the block diagram of the circuit. The system will initially be off by default. The main system will have a converter supplying a load. If the load demands current and power beyond the main converter's capacity. The secondary buck converter will activate.

The system has a negative feedback control structure, where the goal is to regulate the output voltage V_{out} against a reference V_{ref} . The Reference Voltage block represents the desired voltage setpoint. This signal is fed into a Summer, where it is compared against the actual output voltage provided by the Voltage Sensor. The difference between these two signals is the Error Signal.

The PI Controller consists of operational amplifiers in amplifier and integration configuration. Proportional action provides an immediate correction proportional to the current error. Integral action continuously integrates the error over time to eliminate the steady-state error which is the tiny gap that often remains between the target and actual

voltage in simpler systems. The output of the PI controller is a control voltage V_c that dictates how much correction is needed.

The control voltage is then passed to the PWM Generator. This stage converts the analog signal into a high-frequency square wave. By comparing the PI output to a constant-frequency sawtooth wave, the generator varies the Duty Cycle. This PWM is fed into the MOSFET of the converter. The raw output from the MOSFET is a series of high voltage pulses. To convert this into a smooth DC voltage, it passes through the LC filter. The inductor opposes rapid changes in current, smoothing the switching pulses and the capacitor acts as a reservoir, filtering out the remaining voltage ripple to provide a clean DC signal to the load.

The Voltage Sensor monitors the filtered output voltage and feeds it back to the Summing Point. This continuous monitoring ensures that any fluctuations caused by the load or input power are detected and corrected.

3.2 Transfer Function

The LC filter acts as a voltage divider in the frequency domain. The output is connected to a Load Resistor.

Impedance of inductor:

$$Z_L = sL$$

Impedance of output (capacitor || load):

$$Z_{out} = (1/sC) \parallel R = \frac{R}{1 + sRC}$$

We know, the standard second-order system equation is:

$$G_{vd}(s) = \frac{V_{in}}{LC \cdot s^2 + \frac{L}{R} \cdot s + 1}$$

Or in the normalized standard form:

$$G_{vd}(s) = V_{in} \frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$$

Where:

$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ is the resonant frequency.}$$

$Q = R\sqrt{\frac{C}{L}}$ is the quality factor.

At: $L = 1 \text{ mH}$ and $R = 2.5\Omega$

$$\omega_0 = \frac{1}{\sqrt{0.001 \cdot 0.00047}} = 1458 \text{ rad/sec}$$

$$f_0 = \frac{\omega_0}{2\pi} \approx 232 \text{ Hz}$$

The denominator of the transfer function gives us the poles. For an LC filter, these are complex conjugate poles located in the left half plane.

$$s_{1,2} = -\sigma \pm j\omega_d$$

Real Part σ : Controlled by the load R. This provides damping.

Imaginary Part $j\omega_d$: Causes oscillation/ringing.

Because the switching frequency is much higher than the pole frequency, the filter is effective at removing the switching noise. At the resonant frequency $f_0 = 232 \text{ Hz}$, the LC filter introduces a -180° phase shift very rapidly. As inductor lags current by 90° , capacitor lags voltage by 90° . So, a total of 180° .

In a negative feedback loop, if the phase shifts by -180° , the signal inverts. Since negative feedback is already an inversion (-180°), adding another -180° makes it -360° (or 0°).

This turns negative feedback into positive feedback. If the gain is > 1 when this happens, the converter will oscillate uncontrollably and becomes unstable.

But here, the problematic pole lies at 232 Hz. Because 232 Hz is quite low, the gain of the system begins to drop off early. By the time the system reaches 5 kHz, the gain is very low, suppressing noise. The capacitor also introduces a zero into the transfer function. A pole pulls phase down (-90°) and a zero pulls phase up ($+90^\circ$).

We place the PI Controller's zero exactly near the LC Filter's Pole such that the zero cancels out one of the poles. The system remains stable because the total phase shift never reaches the critical positive feedback point while the gain is high.

CHAPTER IV: RESULTS AND DISCUSSIONS

4.1 Simulation of the buck converter

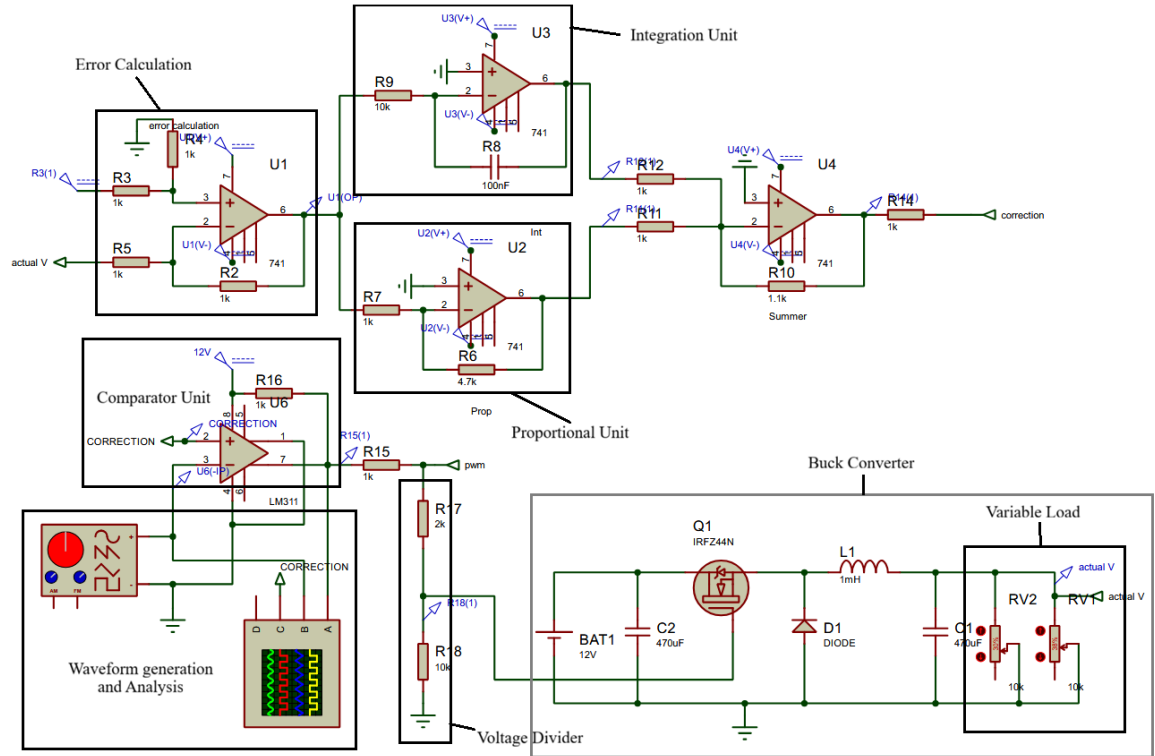


Figure 4.1: Simulation of the buck converter circuit done on Proteus

The simulation circuit is divided into three distinct functional sections: the Control stage, the PWM stage, and the Buck converter.

4.2 Simulation of input signal conditioning

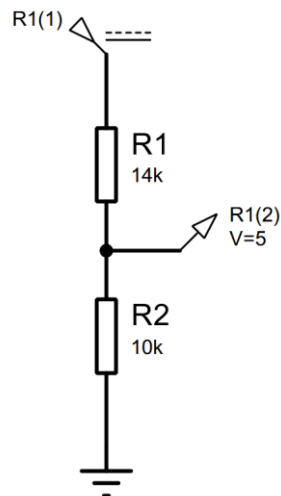


Fig 4.2: Simulation of input signal conditioning

4.3 Simulation of the switch for secondary converter

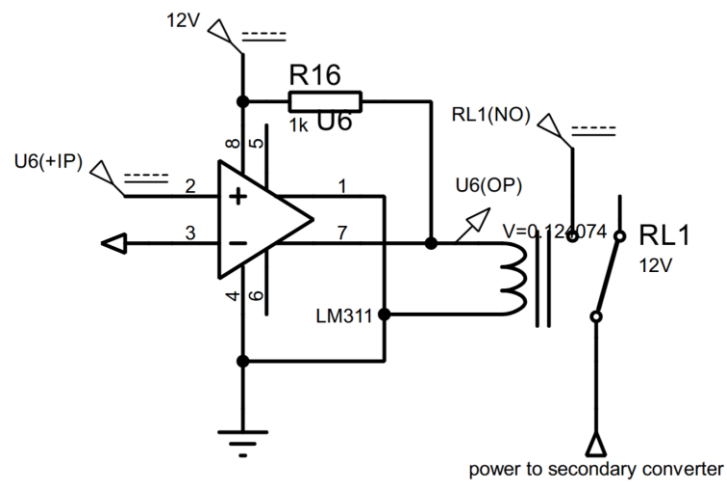


Fig 4.3: Simulation of the switch for secondary converter

The comparator detects a major change in load voltage, if the change is high, the relay is switched, n thus the secondary buck converter is also activated.

4.4 Analysis of output voltage under load fluctuations

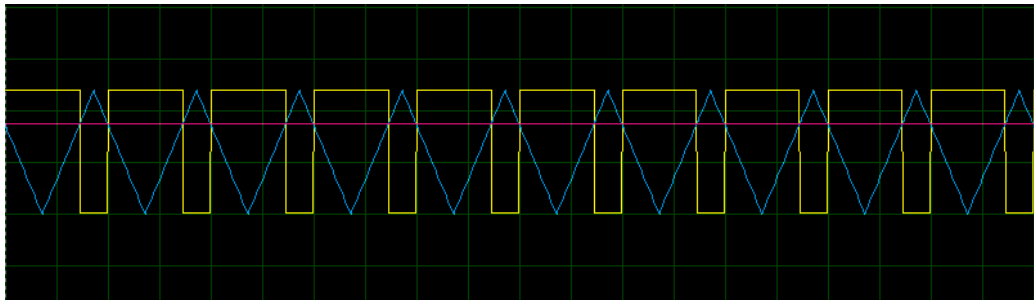


Fig 4.4: Waveform under low load demand

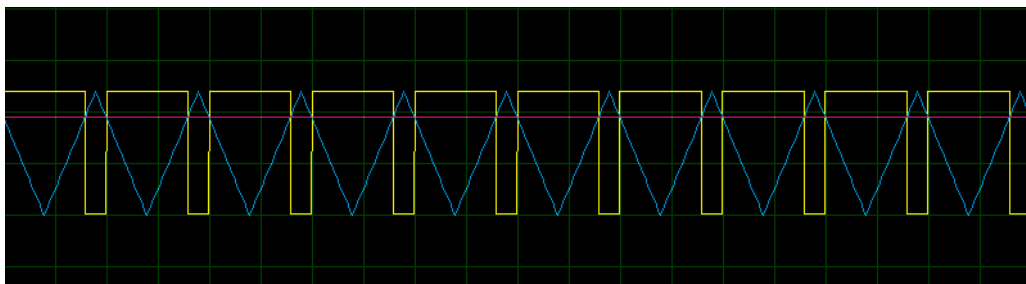


Fig 4.5: Waveform under medium load demand

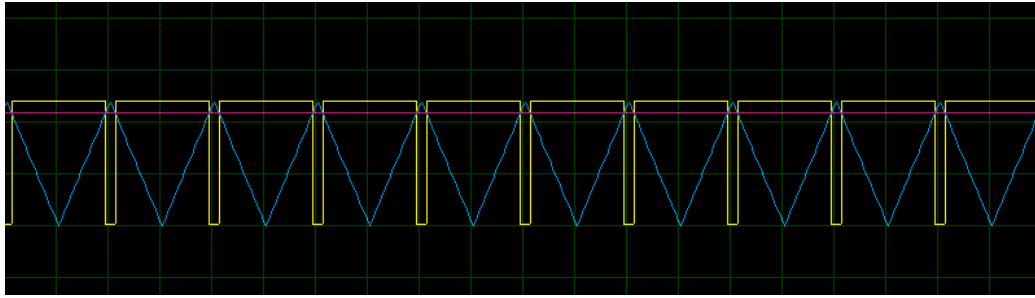


Fig 4.6: Waveform under high load demand

Index:

x-axis Time (s)

y-axis Voltage (V)

In figure 4.4; 4.5, 4.6, blue is the reference triangle wave, red is the error correction signal from the controller and yellow is the generated PWM.

Under varying loads, as the error correction signal changes with the load, the duty cycle also varies accordingly. As the output voltage is directly proportional to the duty cycle, when an undervoltage is seen, the error correction signal increases in magnitude, which increases duty cycle of the PWM, thus the output voltage also increases. Similar process is followed for increased voltage.

CONCLUSION

In conclusion, this project achieved a regulated DC-DC conversion system capable of maintaining a stable output voltage under varying load conditions. The simulation results confirm that the proportional unit provides the necessary speed to react to sudden changes in the load, while the integral unit effectively eliminates the steady-state error. By accumulating the difference between the reference and the actual voltage, the integral action ensures that the output eventually settles exactly at the target value.

The use of the LM311 comparator as a PWM generator proved to be a robust method for translating analog correction signals into high frequency switching signal for the MOSFET. Furthermore, the LC filter successfully smoothed the high frequency switching pulses into a clean DC signal with minimal ripple. The closed-loop system designed in this project effectively addresses the inherent instabilities of an open-loop buck converter. Future iterations of this project could explore the addition of a derivative term to further improve transient response.

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APPENDIX

6.6 Electrical Characteristics

at specified free-air temperature, $V_{CC\pm} = \pm 15\text{ V}$ (unless otherwise noted)

PARAMETER		TEST CONDITIONS		T _A ⁽¹⁾	LM111 LM211 LM211Q			LM311			UNIT
					MIN	TYP ⁽²⁾	MAX	MIN	TYP ⁽²⁾	MAX	
V _{IO}	Input offset voltage	See ⁽³⁾		25°C	0.7 3			2 7.5			mV
				Full range	4			10			
I _{IO}	Input offset current	See ⁽³⁾		25°C	4 10			6 50			nA
				Full range	20			70			
I _{IB}	Input bias current	1 V ≤ V _O ≤ 14 V		25°C	75 100			100 250			nA
				Full range	150			300			
I _{IL(S)}	Low-level strobe current ⁽⁴⁾	V _(strobe) = 0.3 V, V _{ID} ≤ −10 mV		25°C	−3			−3			mA
V _{ICR}	Common-mode input-voltage range ⁽³⁾	Lower range		Full range	−14.7 −14.5			−14.7 −14.5			
		Upper range			13 13.8			13 13.8			
A _{VD}	Large-signal differential-voltage amplification	5 V ≤ V _O ≤ 35 V, R _L = 1 kΩ		25°C	40 200			40 200			V/mV
I _{OH}	High-level (collector) output leakage current	I _(strobe) = −3 mA, V _{ID} = 5 mV		25°C	0.2 10						nA
		V _{OH} = 35 V		Full range	0.5						μA
		V _{ID} = 5 mV, V _{OH} = 35 V		25°C				0.2 50			nA
V _{OL}	Low-level (collector-to-emitter) output voltage	I _{OL} = 50 mA	V _{ID} = −5 mV	25°C	0.75 1.5						V
			V _{ID} = −10 mV	25°C				0.75 1.5			
		V _{CC+} = 4.5 V, V _{CC−} = 0 V, I _{OL} = 8 mA	V _{ID} = −6 mV	Full range	0.23 0.4						
			V _{ID} = −10 mV	Full range				0.23 0.4			
I _{CC+}	Supply current from V _{CC+} output low	V _{ID} = −10 mV,	No load	25°C	5.1 6			5.1 7.5			mA
I _{CC−}	Supply current from V _{CC−} output high	V _{ID} = 10 mV,	No load	25°C	−4.1 −5			−4.1 −5			

Fig: Electrical characteristics of LM311

Electrical Characteristics @ $T_J = 25^\circ\text{C}$ (unless otherwise specified)

Parameter	Min.	Typ.	Max.	Units	Conditions
$V_{(BR)DSS}$	55	—	—	V	$V_{GS} = 0\text{ V}$, $I_D = 250\mu\text{A}$
$\Delta V_{(BR)DSS}/\Delta T_J$	—	0.058	—	V/°C	Reference to 25°C , $I_D = 1\text{ mA}$
$R_{DS(on)}$	—	—	17.5	m Ω	$V_{GS} = 10\text{ V}$, $I_D = 25\text{ A}$ ④
$V_{GS(th)}$	2.0	—	4.0	V	$V_{DS} = V_{GS}$, $I_D = 250\mu\text{A}$
g_{fs}	19	—	—	S	$V_{DS} = 25\text{ V}$, $I_D = 25\text{ A}$ ④
I_{DSS}	Drain-to-Source Leakage Current		25	μA	$V_{DS} = 55\text{ V}$, $V_{GS} = 0\text{ V}$
			250		$V_{DS} = 44\text{ V}$, $V_{GS} = 0\text{ V}$, $T_J = 150^\circ\text{C}$
I_{GSS}	Gate-to-Source Forward Leakage		100	nA	$V_{GS} = 20\text{ V}$
			-100		$V_{GS} = -20\text{ V}$
Q_g	Total Gate Charge		63	nC	$I_D = 25\text{ A}$
Q_{gs}	Gate-to-Source Charge		14		$V_{DS} = 44\text{ V}$
Q_{gd}	Gate-to-Drain ("Miller") Charge		23		$V_{GS} = 10\text{ V}$, See Fig. 6 and 13
$t_{d(on)}$	Turn-On Delay Time		12	ns	$V_{DD} = 28\text{ V}$
t_r	Rise Time		60		$I_D = 25\text{ A}$
$t_{d(off)}$	Turn-Off Delay Time		44		$R_G = 12\Omega$
t_f	Fall Time		45		$V_{GS} = 10\text{ V}$, See Fig. 10 ④
L_D	Internal Drain Inductance		4.5	nH	Between lead, 6mm (0.25in.) from package and center of die contact
L_S	Internal Source Inductance		7.5		
C_{iss}	Input Capacitance		1470	pF	$V_{GS} = 0\text{ V}$
C_{oss}	Output Capacitance		360		$V_{DS} = 25\text{ V}$
C_{rss}	Reverse Transfer Capacitance		88		$f = 1.0\text{ MHz}$, See Fig. 5
E_{AS}	Single Pulse Avalanche Energy②		530⑤ 150⑥	mJ	$I_{AS} = 25\text{ A}$, $L = 0.47\text{ mH}$

Fig: Electrical characteristics of IRFZ44N

Absolute Maximum Ratings

	Parameter	Max.	Units
I_D @ $T_C = 25^\circ\text{C}$	Continuous Drain Current, V_{GS} @ 10V	49	A
I_D @ $T_C = 100^\circ\text{C}$	Continuous Drain Current, V_{GS} @ 10V	35	
I_{DM}	Pulsed Drain Current ①	160	
P_D @ $T_C = 25^\circ\text{C}$	Power Dissipation	94	W
	Linear Derating Factor	0.63	W/ $^\circ\text{C}$
V_{GS}	Gate-to-Source Voltage	± 20	V
I_{AR}	Avalanche Current①	25	A
E_{AR}	Repetitive Avalanche Energy①	9.4	mJ
dv/dt	Peak Diode Recovery dv/dt ③	5.0	V/ns
T_J	Operating Junction and	-55 to + 175	$^\circ\text{C}$
T_{STG}	Storage Temperature Range		
	Soldering Temperature, for 10 seconds		
	Mounting torque, 6-32 or M3 screw	300 (1.6mm from case) 10 lbf•in (1.1N•m)	

Fig: Maximum parameter ratings of IRFZ44N